From the Pulpit to the People: Experimental Evidence on the Impact of Friday Sermons

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Persuasion and the Pulpit

- Long before mass media, persuasion at scale happened via the pulpit.
- Today, sermons reach hundreds of millions weekly (Kramer and Fahmy, 2018).
- Whether religion is persuasive is a fundamental question in social science (Iannaccone, 1998; Iyer, 2016).
  - Does religion **shape** beliefs and behavior?
  - Or does it only **reflect** these things?
- If religious messaging persuades, why:
  - Authority of religious leaders?
  - Religious **arguments** in content?
- Must take into consideration endogenous **supply** (religious leader) and **demand** (congregant) responses.

Research Questions

1. What is the **causal impact** of sermon content on behavior?
2. How do religious vs secular **arguments** impact effectiveness of religious leaders' messaging?
3. How do **supply** and **demand** sides respond endogenously to sermon content?
   a. Supply-Side (religious leaders): Persistence of content
   b. Demand-Side (congregants): Changes in attendance

- Challenges:
  - Religious leaders deciding what to say is non-random.
  - Congregants sort.
  - Artificial settings cannot capture full complexity.

This Project: Randomize Friday Sermon Content

- The Friday sermon: one of the most central religious institutions in Muslim societies.
- Novel partnership with Government of Punjab, Pakistan (some mosques state-run, imams state employees):
  - Randomize mosques to sermons on **women's rights**, **neighbors' rights**, or **business-as-usual** (no state input).
  - Cross-randomize religious vs. secular **arguments**.
  - Randomizing sermon content at scale in its natural setting lets me study **supply** and **demand** responses.
- Sample: universe of 309 state mosques in Punjab attended by **~116,000 people**.
- Outcomes: behavioral measures 2-4 weeks post-treatment; surveys of wives and imams; sermon content and attendance counts in short and long run.

Preview of Results

1. Causal impact of sermons:
   a. **Sermons move behavior**: 0.23 SD (women's rights) and 0.19 SD (neighbors' rights) on preregistered behavioral indices.
   b. **Document indirect effects**: wives of treated men report 0.19 SD more pro-rights beliefs on women, despite never hearing sermons.
2. **Arguments matter, differentially**: religious arguments more effective for women's rights while secular arguments dominate for neighbors' rights.
   → Rationalize with Bayesian updating allowing cross-domain interactions.
3. Endogenous responses:
   a. **Supply-side persistence** in pro-rights content, fades in long run.
   b. **Demand-side backlash** ⇒ lower attendance, persists 6 to 12 months later.
   → Present two-sided theoretical framework that rationalizes this.

Contributions

- Economics of religion: Iannaccone (1998); Clingingsmith, Khwaja and Kremer (2009); Iyer (2016); Bryan, Choi and Karlan (2020); Wang (2021); Buccione (2023); Campbell, Sethuraman and Steffen (2024); Sarkar, Elverdin and Lucek (2026).
  - → **Experimental at-scale estimates of the causal impact of sermon content on behavior; key channel through which religious leaders persuade.**
- Economics of persuasion: DellaVigna and Gentzkow (2010); Haaland, Roth and Wohlfart (2023); Yanagizawa-Drott (2014); La Ferrara, Chong and Duryea (2012).
  - → **Sermons as unique persuasion technology; effect of religious vs secular arguments.**
- Economics of gender and social norms: Jayachandran (2015); Bursztyn, González and Yanagizawa-Drott (2020); Field et al. (2021); Dhar, Jain and Jayachandran (2022); Cullen et al. (2024).
  - → **Effectiveness of religious messaging in changing attitudes about gender.**

Roadmap

Context & Institutional Setting

Experimental Design & Data

Results

Theoretical Framework & Interpretation

Further Results

Conclusion

Context & Institutional Setting

The Friday Prayer in Islam

- *Salat al-Jumu'ah*: most important institution of collective worship in Islam.
- **Attendance obligatory for adult Muslim men.**
- Women largely absent from congregations.



The Friday Sermon

- Prayer preceded by Friday *Khutbah*, delivered by imam.
- Tradition traces to Prophet Muhammad.
- Several states centrally mandate content. In others, imams have full freedom.
- **Scale is difficult to overstate.**
- Existing scholarship on Friday sermons descriptive or correlational.



Setting: Punjab, Pakistan

- 127.6 million people (PBS, 2023).
- Average per capita income: $158 per month (PES, 2026).
- ~98% Muslim:
  - Over 90% report religion is very important in daily life (Pew, 2018).



Religion and Women's Rights in Punjab, Pakistan

- Pakistan ranked last in the WEF Global Gender Gap Report (2025).
- Conservative religious interpretations routinely invoked on women (Weiss 2014).
- However, mainstream Islamic doctrine/history also contain substantial support for women's participation.
- **Foregrounding pro-rights elements lets us test whether religious messaging can shift deeply held beliefs about gender.**



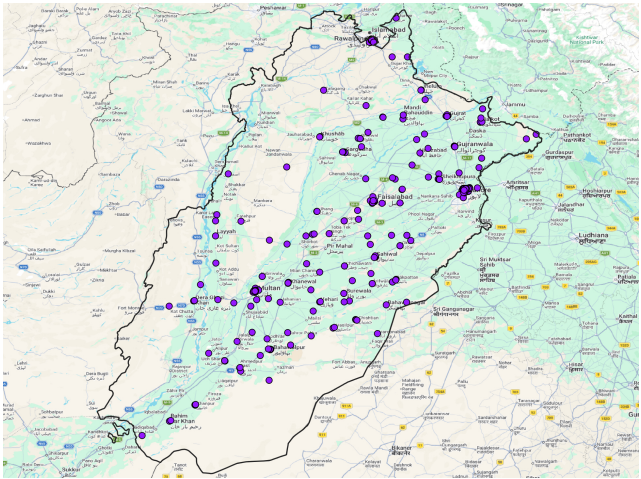
Religion and Neighbors' Rights in Punjab, Pakistan

- Social trust low: only ~23% of Pakistanis agree "most people can be trusted (WVS 2018)."
- Neighborhood public goods under-provided and collective action limited (World Bank 2013).
- Yet Islamic doctrine emphatically pro-neighbor.
- **Unlike gender, conventional religious interpretations reinforce rather than resist pro-rights norms.**



The Auqaf Department and Study Mosques

- Provincial department administering Islamic charitable endowments.
- Directly employs imams.
- **Holds authority to mandate sermon content but does not exercise in normal weeks.**



Experimental Design & Data

Treatments: Two Sermons on Women's or Neighbor's Rights

Women's rights

- Kindness toward women; women's education, employment, and social participation
- Action item: discuss aspirations for daughters with wives

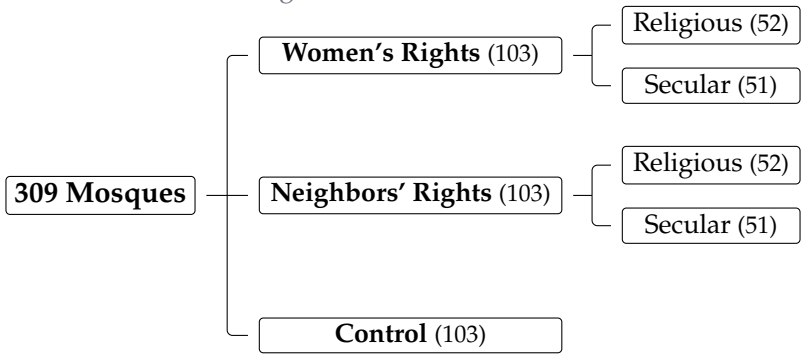
Neighbors' rights

- Kindness toward neighbors; obligation to help neighbors in need; community cleanliness
- Action item: spend an hour cleaning the neighborhood

- **Two distinct scripts** over consecutive Fridays, developed by senior Auqaf religious scholars.
- Test whether religious messaging is more persuasive when it **reinforces** or **challenges** prevailing religious views.

Field Experiment: Design

Figure 1: Randomization



- Randomization stratified by residential/commercial status and attendance tercile.

Treatment Excerpts

	Religious	Secular
Women's Rights	*The Prophet declared women's education just as important as men's education... Rather, the Prophet emphasized the greatness of knowledge in these words: "Seeking knowledge is obligatory upon every Muslim (male and female)."*	*If a woman is educated, she will not only be able to give good advice to her husband in everyday affairs, but she will also be able to participate in the education and upbringing of her children.*
Neighbors' Rights	*The Holy Quran and the Noble Prophet have given great importance to the relationship of neighborliness in their teachings and guidance... The Prophet said: "He is not a believer who sleeps full while knowing his neighbor is hungry."*	*Greeting your neighbors and maintaining good relations with them are the foundations of a peaceful and strong society... Small gestures such as sending each other food and checking on one another in hard times foster unity and brotherhood.*

Example Congregation





Outcomes

- Women's rights outcomes:
  - Dictator game
  - Spouse survey consent
  - Petition
- Neighbors' rights outcomes:
  - Dictator game
  - Community cleanliness
- Effects on wives, imams
- Endogenous responses:
  - Post-treatment sermon content
  - Mosque-level attendance counts



Recruitment

1. Imam introduces study team at Friday prayer as research on religion and social attitudes; sermons never mentioned.
2. Enumerator collects phone numbers of volunteers ($\geq$50 per mosque).
3. Called back in random order; first 15 meeting criteria enrolled:
   - Heard $\geq$10 min of last two sermons in same mosque $\Rightarrow$ regular listeners, likely present at treatment
   - Aged 18–50 and married $\Rightarrow$ daughters' education and employment salient; spouse outcomes feasible

- 71.5% eligible.
- 4,667 men at baseline; 3,892 (84%) at endline; no differential attrition.

Endline Sample: Balanced at Baseline

		Control		Community Treatment		Women Treatment			
	N	Mean	SD	Mean	SD	Mean	SD	Diff1	Diff2
General									
Age	3892	37.67	7.51	37.42	7.60	37.76	7.94	-0.242	0.104
Education	3883	10.00	3.73	9.61	3.81	9.86	3.64	-0.381	-0.121
Income (PKR)	3649	42039.97	24954.99	44297.06	22549.24	42666.11	23506.72	2,250.294	788.294
Daughters Nr	3892	1.31	1.18	1.34	1.23	1.33	1.21	0.028	0.015
Pray (Daily, 0-5)	3881	3.86	1.23	3.95	1.16	3.88	1.23	0.093	0.025
Sermons Attend (Monthly, 0-4)	3886	3.56	0.70	3.52	0.75	3.46	0.77	-0.036	-0.100
Imam Trust (0-10)	3886	8.57	1.80	8.47	1.64	8.53	1.64	-0.100	-0.036
State Trust (0-10)	3861	4.71	3.09	4.72	3.02	4.48	2.89	0.007	-0.225
Neighbors' Rights									
Neighbors Important (0-10)	3890	7.32	2.15	7.37	2.16	7.55	2.19	0.058	0.239
Neighbors Trust (0-10)	3885	7.00	2.13	6.93	2.24	6.94	2.22	-0.069	-0.055
Neighbor Loan (0-10)	3886	5.91	2.39	5.98	2.32	5.80	2.41	0.076	-0.092
Enum Clean (0-10)	3892	6.89	2.16	7.08	2.00	7.07	1.97	0.187	0.183
Belief Index (Neighbors)	3892	0.01	0.55	0.02	0.55	0.01	0.54	0.010	-0.000
Women's Rights									
Men Better Decisions (0-10)	3887	8.12	2.10	7.93	2.08	7.94	2.06	-0.186	-0.176
Women Delay Marriage (0-10)	3868	5.33	2.97	4.82	3.04	5.00	2.95	-0.509*	-0.321
Women Job (0-10)	3883	5.03	2.84	5.03	2.78	5.07	2.82	-0.004	0.041
Spouse Survey (0/1)	3892	0.64	0.48	0.56	0.50	0.60	0.49	-0.076	-0.036
Belief Index (Women)	3892	-0.01	0.43	0.03	0.44	0.03	0.45	0.032	0.040

Notes: * 0.10 ** 0.05 *** 0.01. Standard errors clustered at mosque-level.

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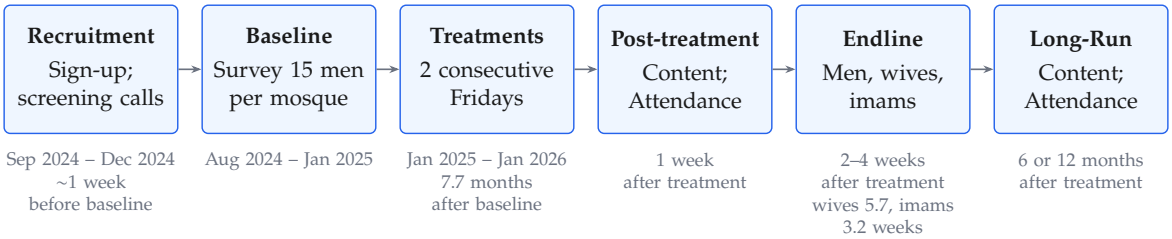
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Timeline



- Treatment and endline rollout staggered district-by-district.
- Treatments delivered ~7.7 months after baseline.
- Outcome measurement (time since treatments):
  - Congregant outcomes: ~3 weeks later
  - Wife outcomes: ~6 weeks later
  - Imam outcomes: ~3weeks later
  - Mosque-level outcomes: 1 week later and 6-12 months later

Estimating Equations

For individual i in mosque m, primary ITT specification:

$$y_{im} = \alpha + \beta_1 \text{WomenRights}_m + \beta_2 \text{NeighborRights}_m + \theta_s + \epsilon_{im}$$

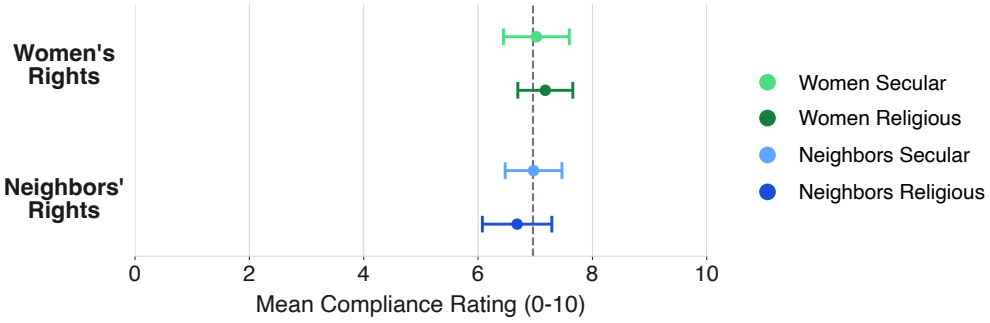
- SEs clustered at mosque level for individual-level outcomes; robust SEs for mosque-level.
- Units of outcomes:
  - Individual-level aggregated into preregistered indices $\Rightarrow \beta$ in SD units.
  - Sermon content $\Rightarrow \beta$ in 0-10 rating points.
  - Attendance $\Rightarrow \beta$ in % changes.

For religious/secular arguments, I disaggregate:

$$\begin{aligned} y_{im} = \alpha &+ \beta_1 \text{WomenReligious}_m + \beta_2 \text{WomenSecular}_m \\ &+ \beta_3 \text{NeighborReligious}_m + \beta_4 \text{NeighborSecular}_m + \theta_s + \epsilon_{im} \end{aligned}$$

and test $\beta_1 = \beta_2$ and $\beta_3 = \beta_4$.

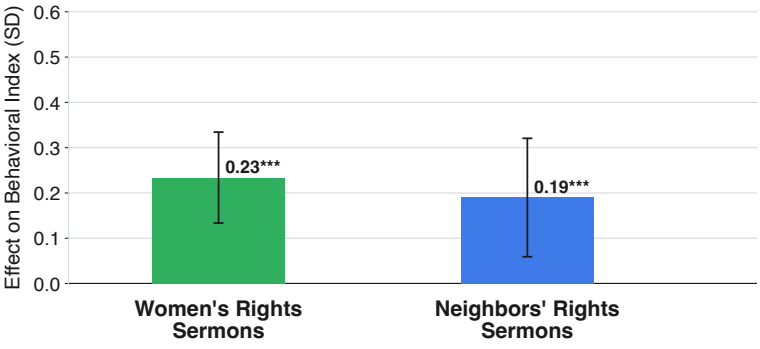
First-Stage: LLM-Rated Similarity of Sermon Content with Script



- Similarity of recorded content to script averaged for both treatment sermons.
- 7/10 average rating for all treatment mosques.
- **No differential compliance across arms.**

Results

Result 1a: Effect of Sermon Content on Behavior of Congregants

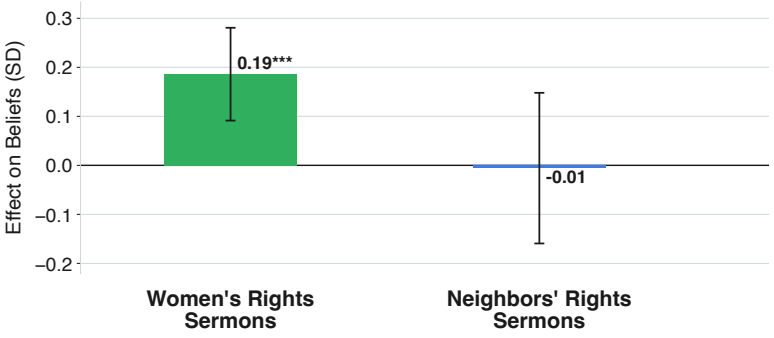


- Two sermons have sizable impacts on behavior.
- Effect sizes comparable to far more intensive interventions.

Notes: N=3,892. SEs clustered at mosque-level. Outcome in green bar is index combining allocation in dictator game to women's employment fund, spouse survey consent, and petition signing. Outcome in blue bar is index combining allocation in dictator game to poor neighbor and enumerator-evaluated community cleanliness.

Table Components Beliefs Self-Reported Behavior

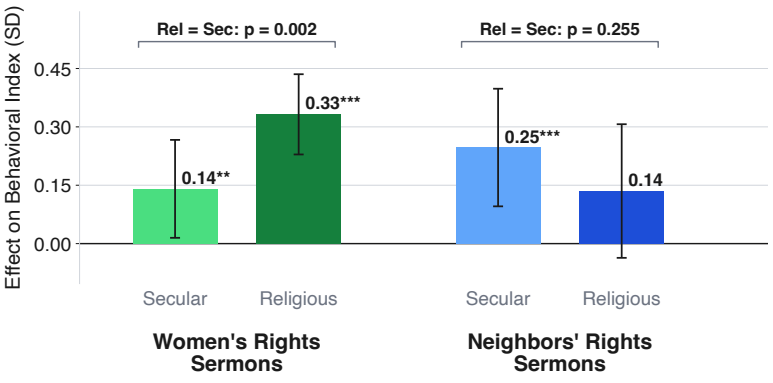
Result 1b: Indirect Effects on Beliefs of Congregants' Wives



- Large impacts on beliefs of wives from women's rights sermons.
- No impact on beliefs of wives from neighbors' rights sermons.

Notes: N=618. SEs clustered at mosque-level. Outcome in green bar is index of beliefs of wives about women's rights. Outcome in blue bar is index of beliefs of wives about neighbors' rights.

Result 2: Effect of Religious vs Secular Arguments

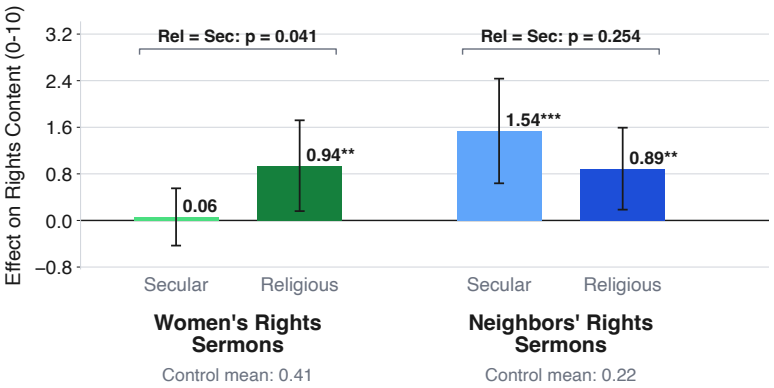


- Religious arguments >2x as effective for women's rights.
- Secular arguments more effective for neighbors' rights.

Notes: N=3,892. SEs clustered at mosque-level. Outcome in green bars is index combining allocation in dictator game to women's employment fund, spouse survey consent, and petition signing. Outcome in blue bars is index combining allocation in dictator game to poor neighbor and enumerator-evaluated community cleanliness.

Table

Result 3a (Supply-Side Responses): Effects on Post-Treatment Sermon Content

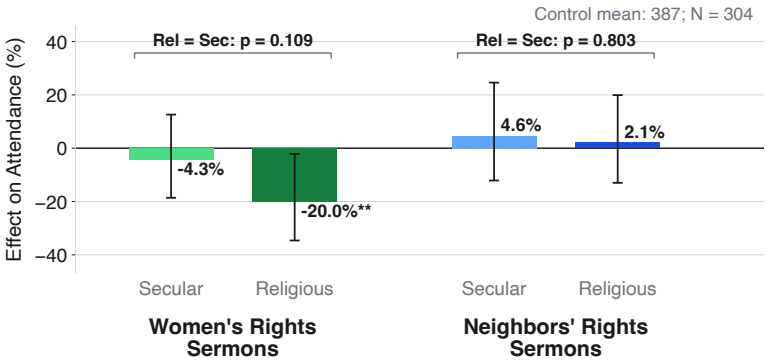


- Imams keep preaching rights content one week later. ⇒**Persistence**.

Notes: N=305 mosques. SEs robust. Outcome is LLM rating (0–10) of women's rights content (green bars) and neighbors' rights content (blue bars) 1 week after conclusion of treatments.

Table | Additional Content Categories

Result 3b (Demand-Side Responses): Effects on Post-Treatment Attendance



- Attendance falls after religious women's rights sermons. ⇒**Backlash**.
- No effect of neighbors' rights sermons.

Notes: N=304 mosques. SEs robust. Outcome is log of enumerator-counted attendance 1 week after conclusion of treatments, shown as percent changes.

Taking Stock

- What we found:
  - Two sermons change behavior, and effects extend indirectly to wives.
  - Type of argument matters, differentially by topic.
  - Imams keep preaching rights.
  - Attendance falls in women-religious mosques.
- Puzzles:
  - What drives differential effects of religious vs secular arguments?
  - Why does the most persuasive arm also lose the most congregants?
  - Why does rights content persist?
  - How would we expect content and attendance to evolve in long run?
- **Framework with demand and supply sides helps makes sense of puzzles.**

Theoretical Framework & Interpretation

Setup: Players

The game consists of, two players, both Bayesian:

- **Imam**: I, chooses **sermon content** C_I each Friday.
- **Congregant**: i, chooses whether to **switch** mosques and how to **behave**, y_i.

Setup: Congregants

- Congregants value a **religious** and a **secular** objective: conforming to God's will (y^*) and doing well in life (z^*). Both uncertain.
- Choose **action** y_i to maximize:

$$-E_i(\alpha\,(y_i - y^*)^2 + (1-\alpha)\,(y_i - z^*)^2 + \kappa_T \Delta_i^2)$$

  where α is religiosity, Δ_i is behavior change sermon induces, and κ_T the stakes of topic T.
- Religious arguments are **signals** about God's will; secular arguments, about worldly payoffs.
- Having heard sermon, congregant must act on new information: cannot shade response to avoid cost.
- Individual i **switches** his mosque if expected cost of revision exceeds benefit by more than switching cost τ:

$$(\kappa_T - 1)E_i(\Delta_i^2) > \tau$$

Setup: Imams

Imam chooses **content** C_I, trading off two motives:

- **Preaching:** utility from teaching what he believes, s_I.
- **Pandering:** utility from closeness to congregants' priors $\bar{s}_{-1}$, inferred from **attendance.**

Imam chooses C_I to maximize:

$$-(C_I - s_I)^2 \; - \; \theta \; (C_I - \bar{s}_{-1})^2$$

Order of Moves

Each Friday:

1. Imam chooses content C_I.
2. Congregants decide whether to switch, given what they expect to hear.
3. Attendees treat sermon as a signal, update beliefs, choose behavior y_i.
4. Imam updates personal beliefs as well as beliefs about congregants' priors (inferred from attendance changes).

The experiment enters at step 1: for two Fridays, the scripts exogenously shift C_I.

Predictions

$$y_i = \underset{(1)}{\alpha E_i(y^*) + (1-\alpha) E_i(z^*)} \qquad \underset{(2)}{(\kappa_T - 1) E_i(\Delta_i^2) > \tau} \qquad C_l = \underset{(3)}{\frac{s_l + \theta \bar{s}_{-1}}{1+\theta}}$$

Demand side:

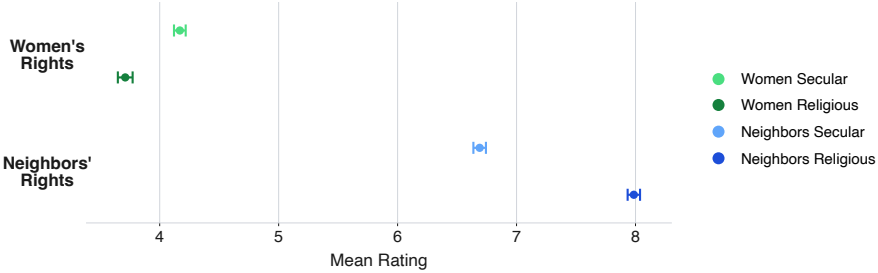
1. Surprise drives persuasion (Bayesian updating): Effects largest where sermon most contradicts priors.
2. No cross-domain interaction: Religious priors irrelevant to how much secular argument persuades, and vice versa.
3. Backlash where revision costly: Attendance falls where induced revision largest and stakes κ_T highest.

Supply side:

1. Sermons make imams more pro-rights $\Rightarrow$ Content becomes more pro-rights.
2. Lower perceived congregant priors $\Rightarrow$ Content becomes less pro-rights.

Further Results

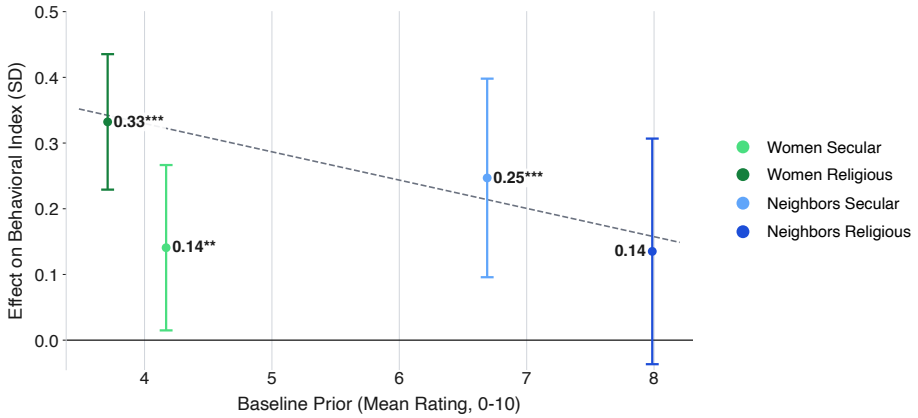
Explaining Demand-Side Results: Religious vs Secular Priors



- Higher ratings (0-10) ⇒ More pro-rights priors.
- Women's rights: religious priors **less** pro-rights than secular.
- Neighbors' rights: religious priors **more** pro-rights than secular.

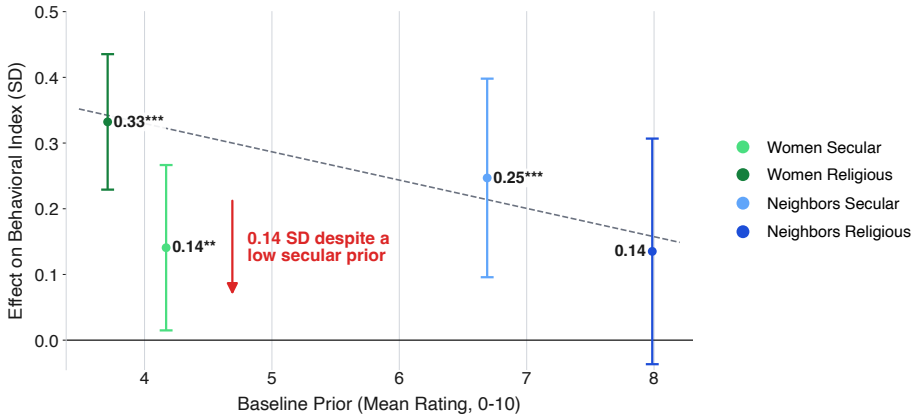
Notes: Means with 95% CIs. Religious prior: mean rating of what Islamic teachings say on each topic (women's decision-making, education, employment; importance of, trust in, obligations toward neighbors). Secular prior: mean of the same questions asked about own beliefs and about community norms.

Explaining Demand-Side Results: Are Mosque-Goers Simply Bayesian?



Notes: SEs clustered at mosque-level. Each dot is an arm's effect size on its behavioral index against the relevant mean baseline prior (0-10).

Explaining Demand-Side Results: Are Mosque-Goers Simply Bayesian?



Notes: SEs clustered at mosque-level. Each dot is an arm's effect size on its behavioral index against the relevant mean baseline prior (0-10).

Modifying Standard Bayesian Framework to Allow Cross-Domain Interactions

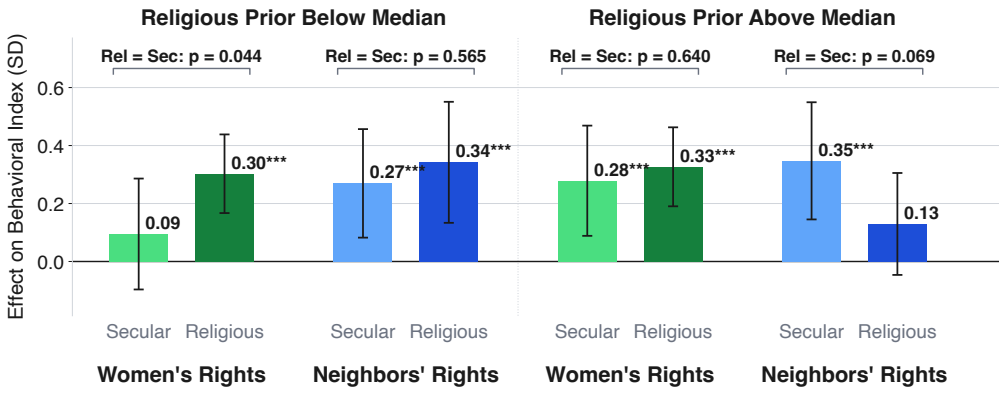
- Needs explaining: Women's rights secular effect small despite low priors.
- Weight secular objective by religious permissibility function $\phi(s_i)$, weakly increasing in religious prior:

$$-E_i\left(\alpha(y_i - y^*)^2 + (1-\alpha)\,\phi(s_i)\,(y_i - z^*)^2\right)$$

  - Low religious priors $\Rightarrow$ under-response to secular signals.
  - High religious priors $\Rightarrow$ over-respond to secular arguments.
  - $\phi = 1$ returns the standard model.

- **New prediction: secular treatment effects increasing in religious priors.**

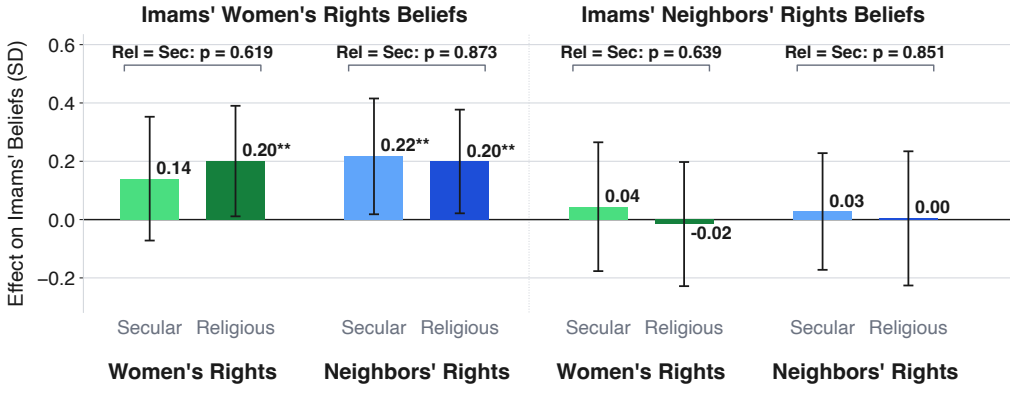
Test: Decomposing Sample by Religious Priors



Notes: N=3,610. SEs clustered at mosque-level. Outcomes are relevant behavioral indices for each topic.

Table

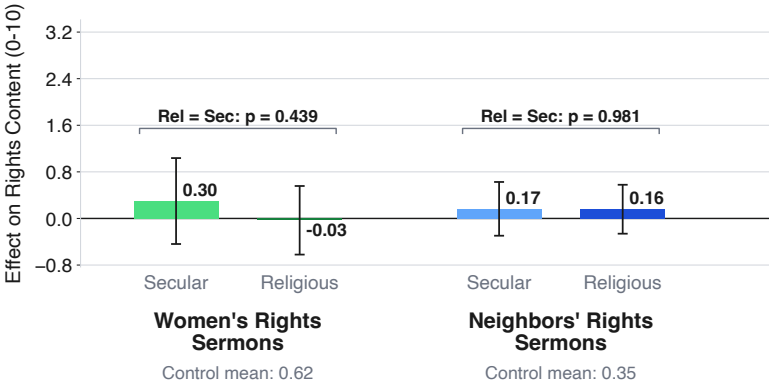
Explaining Supply-Side Results: Imams' Beliefs



Notes: N=242 imams. SEs robust. Outcomes are indices of imams' self-reported beliefs about women's rights (left) and neighbors' rights (right).

Table Job Satisfaction

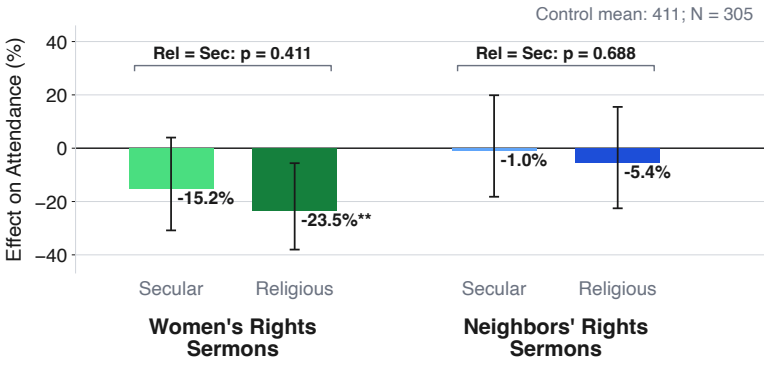
Sermon Content in Long Run



- Rights content decayed in 6–12 months.
- Consistent with pandering.

Notes: N=299 mosques. SEs robust. Outcome is LLM rating (0-10) of women's rights content (green bars) and neighbors' rights content (blue bars) in a sermon recorded 6-12 months after treatment.

Attendance in Long Run



- Attendance effects persist 6-12 months later.
- No change for neighbors' rights mosques.

Notes: N=305 mosques. SEs robust. Outcome is log of enumerator-counted attendance at a Friday prayer 6-12 months after treatment, shown as a percent change.

Table

Conclusion

Conclusion

- Randomize sermon content and explore impacts on behavior.
- Just two sermons moved men's behavior with effects extending indirectly to wives.
- Positive impact of sermons came at a cost $\Rightarrow$ demand-side backlash.
- Present simple framework that rationalizes both persuasion and backlash.
- Results offer evidence for how religious institutions shape economic behavior.

Further Research

- Key policy counterfactual: how would secular leaders fare at delivering the same information?
- Beyond backlash, are there further unintended consequences of utilizing religious leaders for information campaigns?
- What drives religious content in equilibrium and in the long run?
- Political economy of interactions between states and religious leaders and welfare consequences of different sermon regimes.

Thank You!

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Appendix

Appendix: Balance

Table A1: Endline Sample Balance

		Control		Community Treatment		Women Treatment			
	N	Mean	SD	Mean	SD	Mean	SD	Diff1	Diff2
General									
Age	3892	37.67	7.51	37.42	7.60	37.76	7.94	-0.242	0.104
Education	3883	10.00	3.73	9.61	3.81	9.86	3.64	-0.381	-0.121
Income (PKR)	3649	42039.97	24954.99	44297.06	22549.24	42666.11	23506.72	2,250.294	788.294
Daughters Nr	3892	1.31	1.18	1.34	1.23	1.33	1.21	0.028	0.015
Pray (Daily, 0-5)	3881	3.86	1.23	3.95	1.16	3.88	1.23	0.093	0.025
Sermons Attend (Monthly, 0-4)	3886	3.56	0.70	3.52	0.75	3.46	0.77	-0.036	-0.100
Imam Trust (0-10)	3886	8.57	1.80	8.47	1.64	8.53	1.64	-0.100	-0.036
State Trust (0-10)	3861	4.71	3.09	4.72	3.02	4.48	2.89	0.007	-0.225
Neighbors' Rights									
Neighbors Important (0-10)	3890	7.32	2.15	7.37	2.16	7.55	2.19	0.058	0.239
Neighbors Trust (0-10)	3885	7.00	2.13	6.93	2.24	6.94	2.22	-0.069	-0.055
Neighbor Loan (0-10)	3886	5.91	2.39	5.98	2.32	5.80	2.41	0.076	-0.092
Enum Clean (0-10)	3892	6.89	2.16	7.08	2.00	7.07	1.97	0.187	0.183
Belief Index (Neighbors)	3892	0.01	0.55	0.02	0.55	0.01	0.54	0.010	-0.000
Women's Rights									
Men Better Decisions (0-10)	3887	8.12	2.10	7.93	2.08	7.94	2.06	-0.186	-0.176
Women Delay Marriage (0-10)	3868	5.33	2.97	4.82	3.04	5.00	2.95	-0.509*	-0.321
Women Job (0-10)	3883	5.03	2.84	5.03	2.78	5.07	2.82	-0.004	0.041
Spouse Survey (0/1)	3892	0.64	0.48	0.56	0.50	0.60	0.49	-0.076	-0.036
Belief Index (Women)	3892	-0.01	0.43	0.03	0.44	0.03	0.45	0.032	0.040

Notes: * 0.10 ** 0.05 *** 0.01. Standard errors clustered at mosque-level.

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Appendix: Balance - Wives' Sample

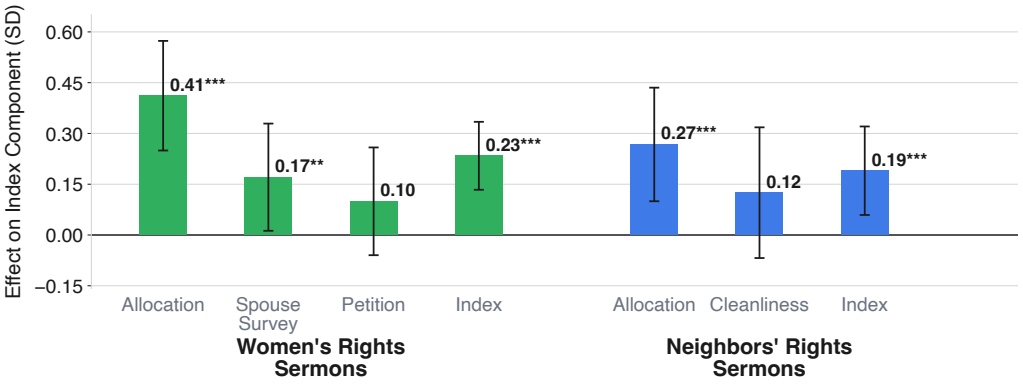
Table A2: Endline Sample Balance (Wives and Husbands)

		Control		Neighbors' Rights		Women's Rights			
	N	Mean	SD	Mean	SD	Mean	SD	Diff1	Diff2
Baseline Wives Sample									
Age	113	33.88	9.08	34.69	9.32	34.82	6.20	0.812	0.937
Education	113	8.61	4.62	8.57	4.90	7.29	5.00	-0.035	-1.317
Job (0/1)	113	0.00	0.00	0.10	0.30	0.05	0.23	0.095	0.053
Job Search (0/1)	113	0.06	0.24	0.05	0.22	0.00	0.00	-0.013	-0.061
Children Nr	113	3.15	1.79	3.36	2.14	3.00	1.59	0.206	-0.152
Daily Pray (0-5)	113	4.09	1.28	4.07	1.35	4.08	1.10	-0.019	-0.012
Well-Being	113	0.00	0.83	0.10	0.65	-0.04	0.75	0.098	-0.050
Beliefs Women	113	0.04	0.50	0.05	0.50	0.09	0.32	0.009	0.051
Beliefs Neighbors	113	-0.01	0.56	0.04	0.54	-0.04	0.44	0.048	-0.032
Full Wives Sample									
Age (Husband)	618	37.97	7.14	37.07	7.32	37.65	7.76	-0.904	-0.321
Education (Husband)	618	10.08	3.60	9.33	3.54	10.01	3.54	-0.757	-0.069
Income (Husband, PKR)	585	42807.02	25000.31	42000.00	23371.35	40931.37	21624.09	-807.018	-1,875.645
Children Nr (Husband)	616	2.82	1.63	2.80	1.74	2.57	1.66	-0.021	-0.249
Daily Pray (Husband, 0-5)	613	3.81	1.25	3.92	1.20	3.83	1.31	0.112	0.022
Beliefs Women (Husband)	618	0.03	0.40	0.03	0.45	0.05	0.42	-0.002	0.014
Beliefs Neighbors (Husband)	618	0.01	0.56	0.07	0.57	-0.01	0.51	0.062	-0.017
Spouse Survey (Husband, 0/1)	618	0.70	0.46	0.67	0.47	0.58	0.50	-0.029	-0.125
Enum Clean (Husband, 0-10)	618	6.77	2.12	6.96	1.93	6.83	2.19	0.184	0.051

Notes: * 0.10 ** 0.05 *** 0.01. SEs clustered at mosque-level. Upper panel: the ~19% of endline wives matched to a baseline wife. Lower panel: husbands' baseline characteristics, full endline wives' sample.

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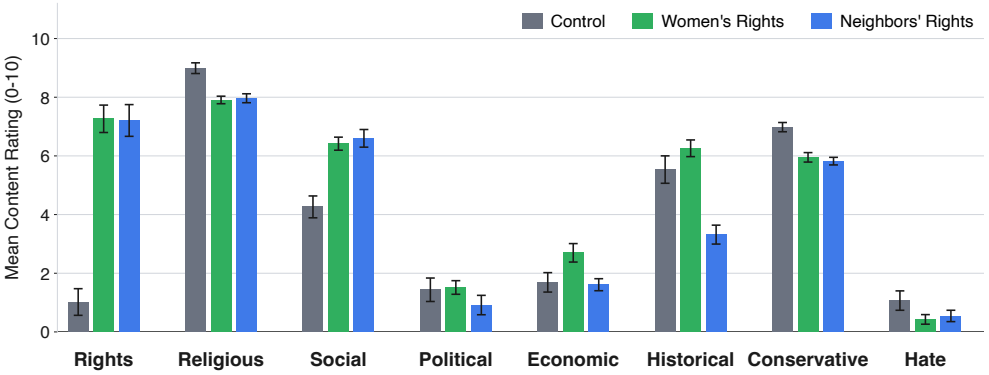
Appendix: Components of the Behavioral Indices



Notes: N=3,892. SEs clustered at mosque-level. Bars are standardized components of the behavioral indices: allocation in dictator game to women's employment fund, spouse survey consent, and petition signing (green); allocation in dictator game to poor neighbor and enumerator-evaluated community cleanliness (blue).

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Results: Sermon Content During Treatment Weeks



Notes: N=103 control, 102 women's rights, 101 neighbors' rights mosques. Means with 95% CIs. Bars are LLM ratings (0-10) of eight content categories in sermons delivered during the treatment weeks: average of the two treatment sermons in treated mosques, first recorded sermon in control mosques.

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Appendix: Beliefs

Table A3: Beliefs about Women and Neighbors

Dependent Variable	Beliefs about Women				Beliefs about Neighbors			
	Competence	Education	Employment	Pooled	Importance	Trust	Help	Pooled
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Women's Rights	0.0660	0.1758***	0.1135	0.1149***	-0.1149	-0.0468	-0.1352**	-0.0867*
	(0.0534)	(0.0637)	(0.0749)	(0.0438)	(0.0708)	(0.0540)	(0.0593)	(0.0471)
Neighbors' Rights	-0.0240	0.0761	0.0356	0.0479	0.1034	0.0546	0.0680	0.0560
	(0.0481)	(0.0615)	(0.0723)	(0.0395)	(0.0723)	(0.0560)	(0.0676)	(0.0575)
N	3892	3892	3892	3892	3892	3892	3892	3892
R-Squared	0.0485	0.0919	0.0457	0.0894	0.0702	0.0555	0.0784	0.0691
Control Mean	0.0003	-0.0009	0.0018	-0.0004	-0.0019	0.0002	0.0004	0.0006

Notes: * 0.10 ** 0.05 *** 0.01. Standard errors clustered at mosque-level. Each column is an index created following Anderson (2008) from responses to multiple questions; the pooled column pools the preceding three columns into one index.

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Appendix: Table - Self-Reported Behavior (Women's Rights)

Table A4: Self-Reported Behavior (Women's Rights)

Dependent Variable	Time-Use (Daily Hrs)			Wife			
	Wife	Daughters	Sons	Job Hrs	Chores Hrs	Outside Hrs	Comfort
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Women's Rights	-0.0049	-0.1439	0.0285	-0.2019	-0.4135*	-0.0172	0.3525**
	(0.2351)	(0.2087)	(0.2087)	(0.1824)	(0.2173)	(0.1965)	(0.1729)
Neighbors' Rights	0.0720	-0.1432	-0.0611	-0.2744	0.0244	0.1585	0.1527
	(0.2425)	(0.2109)	(0.2069)	(0.1863)	(0.2103)	(0.1921)	(0.1754)
N	3841	2650	2822	3892	3892	3892	3873
R-Squared	0.0152	0.0222	0.0199	0.0324	0.0272	0.0136	0.0434
Control Mean	3.4373	2.9026	2.8089	1.2764	5.3048	2.6251	7.4124

Notes: * 0.10 ** 0.05 *** 0.01. Standard errors clustered at mosque-level. Columns (1)-(3) are the respondent's self-reported daily hours talking with his wife, daughters, and sons; (4)-(6) are his estimates of his wife's daily hours in paid work, on chores, and outside the home; (7) his comfort talking with his wife (0-10).

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Appendix: Table - Self-Reported Behavior (Neighbors' Rights)

Table A5: Self-Reported Behavior (Neighbors' Rights)

Dependent Variable	Time-Use (Weekly Hrs)		Neighbors			
	Neighbors	Improving	Know	Ask Help	Receive Help	Food Send
	(1)	(2)	(3)	(4)	(5)	(6)
Women's Rights	-0.0595	-0.0955	1.2855*	-0.2117	-0.1520	-0.6373**
	(0.1780)	(0.1371)	(0.7389)	(0.2224)	(0.2178)	(0.2847)
Neighbors' Rights	0.4367**	0.5072***	2.1451***	0.5289*	0.6386**	0.5086
	(0.1942)	(0.1654)	(0.6609)	(0.2724)	(0.2828)	(0.4126)
N	3892	3892	3892	3892	3892	3892
R-Squared	0.0726	0.0644	0.1217	0.0742	0.0801	0.1253
Control Mean	2.3010	1.3218	6.7182	1.7806	2.0608	3.5119

Notes: * 0.10 ** 0.05 *** 0.01. Standard errors clustered at mosque-level. Columns (1)-(2) are self-reported weekly hours interacting with neighbors and improving the community; (3) the number of neighbors known by name; (4)-(6) the number of times in the past month neighbors asked for help, the respondent received help, and his household sent food to neighbors.

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Appendix: Table - Prosocial Beliefs

Table A6: Prosocial Beliefs

Dependent Variable	**Women's Rights**	**Neighbors' Rights**	**Overall**
	(1)	(2)	(3)
Women's Rights	0.1149***	-0.0867*	0.0318
	(0.0438)	(0.0471)	(0.0301)
Neighbors' Rights	0.0479	0.0560	0.0497
	(0.0395)	(0.0575)	(0.0304)
N	3892	3892	3892
R-Squared	0.0894	0.0691	0.0735
Control Mean	-0.0004	0.0006	0.0012

Notes: * 0.10 ** 0.05 *** 0.01. Standard errors clustered at mosque-level. Each column is an index created following Anderson (2008): beliefs about women's rights (1), about neighbors' rights (2), and all belief questions pooled (3).

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Appendix: Table - Trust

Table A7: Trust

Dependent Variable	Imam Trust				State
	Overall	Voting	Religious	Worldly	Overall
	(1)	(2)	(3)	(4)	(5)
Women Secular	0.2782	-0.0349	0.2040	-0.0384	0.5826**
	(0.2214)	(0.4145)	(0.1854)	(0.3835)	(0.2481)
Women Religious	0.0352	0.1121	-0.0818	0.1493	0.0276
	(0.2374)	(0.4243)	(0.1933)	(0.3333)	(0.2447)
Neighbors Secular	0.0514	0.2671	0.0138	0.2855	0.1035
	(0.2599)	(0.3843)	(0.2073)	(0.3271)	(0.2456)
Neighbors Religious	0.2756	0.1004	0.0951	0.1503	0.1061
	(0.2421)	(0.3755)	(0.1935)	(0.3324)	(0.2777)
N	3882	3854	3885	3874	3837
R-Squared	0.0391	0.0570	0.0350	0.0351	0.0925
Control Mean	7.8128	5.3624	8.1278	6.3840	5.3841
Ho: Women Rel = Women Sec	0.3489	0.7699	0.2036	0.6666	0.0617
Ho: Neighbors Rel = Neighbors Sec	0.4491	0.7007	0.7363	0.7245	0.9936

Notes: * 0.10 ** 0.05 *** 0.01. Standard errors clustered at mosque-level. All outcomes are 0-10 ratings: trust in own imam overall (1), for guidance on voting (2), on religious (3) and on worldly matters (4); trust in state institutions (5).

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Appendix: Additional Content Categories

Table A8: Post-Treatment Sermon Content

Dependent Variable	**Religious**	**Social**	**Political**	**Economic**	**Historical**	**Conservative**	**Hate**
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Women Secular	0.1479	-1.0420***	-0.8041*	-0.5004	-0.8176*	-0.6656***	-1.0185***
	(0.2410)	(0.3379)	(0.4100)	(0.3279)	(0.4243)	(0.1470)	(0.2592)
Women Religious	0.0135	0.0968	-0.4897	-0.2875	-0.3853	-0.5667***	-0.9148***
	(0.2148)	(0.3561)	(0.4411)	(0.3487)	(0.3456)	(0.1393)	(0.2798)
Neighbors Secular	-0.0569	-0.2799	-0.2985	-0.3208	-1.3577***	-0.5367***	-0.4332
	(0.2027)	(0.3615)	(0.4689)	(0.2977)	(0.4166)	(0.1559)	(0.3579)
Neighbors Religious	0.0525	-0.6719*	-0.5218	-0.6853**	-0.8907**	-0.7073***	-0.5626
	(0.2043)	(0.3826)	(0.4802)	(0.2804)	(0.4103)	(0.1542)	(0.3420)
N	305	305	305	305	305	305	305
R-Squared	0.0130	0.0501	0.0168	0.0314	0.0508	0.1140	0.0500
Control Mean	8.7549	4.6176	2.2157	1.9608	5.7059	7.1569	1.5294
Ho: Women Rel = Women Sec	0.6095	0.0038	0.4913	0.5979	0.3634	0.5713	0.6265
Ho: Neighbors Rel = Neighbors Sec	0.6210	0.3655	0.6821	0.2513	0.3685	0.3777	0.7337

Notes: * 0.10 ** 0.05 *** 0.01. Standard errors are robust. Each column is a 0–10 LLM content rating of the first post-treatment sermon.

Appendix: Additional Content Categories in the Long Run

Table A9: Sermon Content 6-12 months After the Intervention

Dependent Variable	**Religious**	**Social**	**Political**	**Economic**	**Historical**	**Conservative**	**Hate**
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Women Secular	0.1239	-0.6882*	-0.0291	-0.6997*	0.5316	0.0010	-0.1249
	(0.1113)	(0.3990)	(0.3498)	(0.3885)	(0.4538)	(0.1589)	(0.3515)
Women Religious	0.1924**	-0.4937	-0.5847**	-0.6004	-0.2846	-0.0034	-0.5989**
	(0.0970)	(0.4449)	(0.2859)	(0.4088)	(0.4258)	(0.1515)	(0.2811)
Neighbors Secular	0.1668*	-0.4522	-0.5923**	-0.2565	0.4881	-0.0531	-0.3634
	(0.0946)	(0.4295)	(0.2657)	(0.4370)	(0.4579)	(0.1357)	(0.2973)
Neighbors Religious	-0.1703	-0.2551	-0.1468	-0.4294	-0.0341	0.0286	0.1960
	(0.2212)	(0.4224)	(0.3292)	(0.4092)	(0.5010)	(0.1301)	(0.3391)
N	299	299	299	299	299	299	299
R-Squared	0.0399	0.0295	0.0404	0.0428	0.0268	0.0174	0.0292
Control Mean	9.8137	4.7745	1.2255	2.7059	5.8824	7.8922	1.0784
Ho: Women Rel = Women Sec	0.2820	0.6974	0.1253	0.8101	0.0923	0.9804	0.1746
Ho: Neighbors Rel = Neighbors Sec	0.0999	0.6962	0.1701	0.7070	0.3475	0.5547	0.1103

Notes: * 0.10 ** 0.05 *** 0.01. Standard errors are robust. Each column is a 0–10 LLM content rating of the sermon recorded 6–12 months after the intervention.

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Appendix: Deriving the Treatment Effects (1/2)

- Congregant i chooses y_i to maximize

$$-E_i(\alpha\,(y_i - y^*)^2 + (1-\alpha)\,(y_i - z^*)^2 + \kappa_T \Delta_i^2) \qquad \Rightarrow \qquad y_i \;=\; \alpha\, E_i(y^*) + (1-\alpha)\, E_i(z^*)$$

- Priors: $s_i \sim N(y^*, 1/p_y)$ and $z_i \sim N(z^*, 1/p_z)$. Signals: $R_i \sim N(y^*, 1/p_R)$ from a religiously framed sermon, $N_i \sim N(z^*, 1/p_N)$ from a secularly framed one.

- Normal priors and signals, posterior means are precision-weighted averages:

$$E_i(y^* \mid R_i) \;=\; \frac{p_y\, s_i + p_R\, R_i}{p_y + p_R} \qquad\qquad E_i(z^* \mid N_i) \;=\; \frac{p_z\, z_i + p_N\, N_i}{p_z + p_N}$$

- A religious signal moves only $E_i(y^*)$; a secular signal moves only $E_i(z^*)$.

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Appendix: Deriving the Treatment Effects (2/2)

- Control (no signal): $y_i^C = \alpha\, s_i + (1-\alpha)\, z_i$
- Treated, optimal y_i:

$$y_i^{Rel} = \alpha \frac{p_y\, s_i + p_R\, R_i}{p_y + p_R} + (1-\alpha)\, z_i \qquad y_i^{Sec} = \alpha\, s_i + (1-\alpha) \frac{p_z\, z_i + p_N\, N_i}{p_z + p_N}$$

- Subtracting y_i^C:

$$\Delta_i^{Rel} = y_i^{Rel} - y_i^C = \alpha \left(\frac{p_y\, s_i + p_R\, R_i}{p_y + p_R} - s_i \right) = \alpha \frac{p_R}{p_y + p_R} (R_i - s_i)$$

$$\Delta_i^{Sec} = y_i^{Sec} - y_i^C = (1-\alpha) \left(\frac{p_z\, z_i + p_N\, N_i}{p_z + p_N} - z_i \right) = (1-\alpha) \frac{p_N}{p_z + p_N} (N_i - z_i)$$

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Appendix: Extension - Correlated Priors (1/2)

- Extension: congregant i's religious and secular priors have correlated errors, $\rho \in (-1, 1)$:

$$\begin{pmatrix} s_i \\ z_i \end{pmatrix} \sim N\left(\begin{pmatrix} y^* \\ z^* \end{pmatrix}, \begin{pmatrix} 1/p_y & \rho/\sqrt{p_y p_z} \\ \rho/\sqrt{p_y p_z} & 1/p_z \end{pmatrix} \right) \qquad R_i,\ N_i \text{ as before}$$

- Action rule and y_i^C unchanged. Own-domain updating unchanged. Signal now also moves the other belief:

$$E_i(z^* \mid N_i) = z_i + \tfrac{p_N}{p_z + p_N}(N_i - z_i) \qquad E_i(y^* \mid N_i) = s_i + \rho\sqrt{\tfrac{p_z}{p_y}}\,\tfrac{p_N}{p_z + p_N}(N_i - z_i)$$

$$E_i(y^* \mid R_i) = s_i + \tfrac{p_R}{p_y + p_R}(R_i - s_i) \qquad E_i(z^* \mid R_i) = z_i + \rho\sqrt{\tfrac{p_y}{p_z}}\,\tfrac{p_R}{p_y + p_R}(R_i - s_i)$$

- Cross-domain update = own-domain update $\times\ \rho\sqrt{p_z/p_y}$.

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Appendix: Extension - Correlated Priors (2/2)

- Plug posteriors into the action rule and subtract y_i^C:

$$\Delta_i^{Rel} = \left[\alpha + (1-\alpha)\,\rho\sqrt{\frac{p_y}{p_z}}\right]\frac{p_R}{p_y + p_R}\,(R_i - s_i)$$

$$\Delta_i^{Sec} = \left[(1-\alpha) + \alpha\,\rho\sqrt{\frac{p_z}{p_y}}\right]\frac{p_N}{p_z + p_N}\,(N_i - z_i)$$

- $\rho = 0$ returns the standard model.
- Still no cross-domain interaction.

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Appendix: Deriving the Gated Treatment Effects (1/2)

- Congregant i maximizes:

$$-E_i\left(\alpha\,(y_i - y^*)^2 + (1-\alpha)\,\phi(s_i)\,(y_i - z^*)^2\right)$$

- First-order condition:

$$y_i \;=\; \frac{\alpha\,E_i(y^*) + (1-\alpha)\,\phi(s_i)\,E_i(z^*)}{\alpha + (1-\alpha)\,\phi(s_i)}$$

- Effective religiosity, weakly decreasing in the religious prior s_i:

$$\tilde{\alpha}_i \;\equiv\; \frac{\alpha}{\alpha + (1-\alpha)\,\phi(s_i)} \qquad \Rightarrow \qquad y_i \;=\; \tilde{\alpha}_i\,E_i(y^*) + (1-\tilde{\alpha}_i)\,E_i(z^*)$$

Appendix: Deriving the Gated Treatment Effects (2/2)

- Control: $y_i^C = \tilde{\alpha}_i\, s_i + (1 - \tilde{\alpha}_i)\, z_i$
- Treated:

$$y_i^{Rel} = \tilde{\alpha}_i \frac{p_y\, s_i + p_R\, R_i}{p_y + p_R} + (1 - \tilde{\alpha}_i)\, z_i \qquad y_i^{Sec} = \tilde{\alpha}_i\, s_i + (1 - \tilde{\alpha}_i) \frac{p_z\, z_i + p_N\, N_i}{p_z + p_N}$$

- Subtracting y_i^C:

$$\Delta_i^{Rel} = \tilde{\alpha}_i \frac{p_R}{p_y + p_R} (R_i - s_i) = \frac{\alpha}{\alpha + (1 - \alpha)\, \phi(s_i)} \frac{p_R}{p_y + p_R} (R_i - s_i)$$

$$\Delta_i^{Sec} = (1 - \tilde{\alpha}_i) \frac{p_N}{p_z + p_N} (N_i - z_i) = \frac{(1 - \alpha)\, \phi(s_i)}{\alpha + (1 - \alpha)\, \phi(s_i)} \frac{p_N}{p_z + p_N} (N_i - z_i)$$

- $\phi \equiv 1$ returns the standard model.
- ϕ increasing $\Rightarrow \Delta_i^{Sec}$ rises and Δ_i^{Rel} falls with the religious prior s_i, beyond signal surprise.

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Appendix: Table - Behavioral Outcomes

Table A10: Primary Congregant Outcomes

Dependent Variable	Women's Rights				Neighbors' Rights		
	Allocation	**Spouse Survey**	**Petition**	**Index**	**Allocation**	**Cleanliness**	**Index**
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Women's Rights	37.0819***	0.0846**	0.0494	0.2340***	0.9456	0.1408	0.0559
	(7.4395)	(0.0400)	(0.0403)	(0.0512)	(7.2097)	(0.1585)	(0.0660)
Neighbors' Rights	6.2484	0.0425	-0.0230	0.0369	24.0638***	0.1873	0.1899***
	(7.1849)	(0.0429)	(0.0395)	(0.0576)	(7.7036)	(0.1476)	(0.0667)
N	3892	3892	3892	3892	3892	3892	3892
R-Squared	0.0334	0.0054	0.0102	0.0292	0.0285	0.0037	0.0179
Control Mean	111.5473	0.4326	0.4380	-0.0003	127.0208	7.0485	0.0019

Notes: * 0.10 ** 0.05 *** 0.01. Standard errors clustered at mosque-level. Index created following Anderson (2008).

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Appendix: Table - Beliefs of Wives

Table A11: Beliefs of Wives

Dependent Variable	**Women's Rights Beliefs**	**Neighbors' Rights Beliefs**
	(1)	(2)
Women's Rights	0.1860***	0.0780
	(0.0483)	(0.0753)
Neighbors' Rights	0.1191**	-0.0057
	(0.0502)	(0.0784)
N	618	618
R-Squared	0.0419	0.0088
Control Mean	-0.0118	0.0091

Notes: * 0.10 ** 0.05 *** 0.01. Standard errors clustered at mosque-level. Each column is an index of beliefs of congregants' wives' created following Anderson (2008).

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Appendix: Table - Employment Behavior of Wives

Table A12: Employment Behavior of Wives

Dependent Variable	**Job**	**Job Search**	**Job Search Easy**
	(1)	(2)	(3)
Women's Rights	-0.0005	-0.0220	-0.4769
	(0.0316)	(0.0257)	(0.3512)
Neighbors' Rights	-0.0037	-0.0360	-0.2851
	(0.0267)	(0.0239)	(0.3341)
N	618	618	618
R-Squared	0.0051	0.0202	0.0204
Control Mean	0.0769	0.0769	4.8791

Notes: * 0.10 ** 0.05 *** 0.01. Respondents are congregants' wives. Standard errors clustered at mosque-level. Outcomes: wife has a paid job (1), is searching for a job (2), rates how easy it would be to find one (3, 0-10).

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Appendix: Table - Perceptions of Husbands' Time-Use and Beliefs

Table A13: Perceptions of Husbands' Time-Use and Beliefs

Dependent Variable	Husbands' Time-Use			Husbands' Beliefs	
	Wife	Sons	Daughters	Daughter Edu	Daughter Job
	(1)	(2)	(3)	(4)	(5)
Women's Rights	-1.2358***	-1.2211***	-1.0392***	0.7659**	-0.2799
	(0.3966)	(0.3960)	(0.3376)	(0.3743)	(0.2898)
Neighbors' Rights	-0.6847	-0.5863	-0.6322*	0.1685	-0.2218
	(0.4148)	(0.4076)	(0.3561)	(0.3812)	(0.3146)
N	618	487	462	618	618
R-Squared	0.0467	0.0451	0.0425	0.0296	0.0084
Control Mean	4.4176	3.9653	3.7574	12.7363	5.3901

Notes: * 0.10 ** 0.05 *** 0.01. Respondents are congregants' wives. Standard errors clustered at mosque-level. Columns (1)-(3) are the wife's report of daily hours her husband spends talking with her, with sons, and with daughters; (4) his desired years of schooling for daughters; (5) whether he would like their daughter to have a job.

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Appendix: Table - Religious vs Secular Framing

Table A14: Religious vs Secular Framing

Dependent Variable	**Women's Rights Index**	**Neighbors' Rights Index**
	(1)	(2)
Women Secular	0.1407**	0.0279
	(0.0642)	(0.0858)
Women Religious	0.3322***	0.0855
	(0.0526)	(0.0761)
Neighbors Secular	0.1189	0.2469***
	(0.0735)	(0.0771)
Neighbors Religious	-0.0416	0.1351
	(0.0677)	(0.0876)
N	3892	3892
R-Squared	0.0417	0.0205
Control Mean	-0.0003	0.0019
Ho: Women Rel = Women Sec	0.0020	0.5446
Ho: Neighbors Rel = Neighbors Sec	0.0546	0.2550

Notes: * 0.10 ** 0.05 *** 0.01. Standard errors clustered at mosque-level. Outcomes are primary behavioral indices, created following Anderson (2008); p-values at the foot test equality of the religious and secular arms.

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Appendix: Heterogeneity by Baseline Priors

Table A15: Heterogeneity by Baseline Religious and Secular Priors

Dependent Variable	**Women's Rights**	**Neighbors' Rights**
	(1)	(2)
Religious	0.3027***	0.3421***
	(0.0692)	(0.1065)
Secular	0.0948	0.2695***
	(0.0976)	(0.0955)
Religous x	0.0240	-0.2126**
1(Religious Prior > Median)	(0.0765)	(0.1046)
Secular x	0.1838*	0.0778
1(Religious Prior > Median)	(0.0991)	(0.1093)
Religious x	-0.0117	-0.2222**
1(Secular Prior > Median)	(0.0712)	(0.1041)
Secular x 1(Secular	-0.1176	-0.1935*
Prior > Median)	(0.1034)	(0.0999)
N	3610	3610
R-Squared	0.0406	0.0446
Control Mean	-0.0055	-0.0006

Notes: * 0.10 ** 0.05 *** 0.01. Standard errors clustered at mosque-level. Outcomes are primary behavioral indices, created following Anderson (2008); priors are baseline 0-10 ratings split at the median.

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Appendix: Table - Personal vs Norms vs Religious Beliefs

Table A16: Personal vs Norms vs Religious Beliefs

Dependent Variable	Women's Rights			Neighbors' Rights		
	Personal	**Norms**	**Religious**	**Personal**	**Norms**	**Religious**
	(1)	(2)	(3)	(4)	(5)	(6)
Women's Rights	0.0711	0.0905	0.0974	-0.1174*	-0.1528**	0.0441
	(0.0688)	(0.0670)	(0.0845)	(0.0665)	(0.0731)	(0.0938)
Neighbors' Rights	-0.0506	-0.0642	-0.0281	0.1067	0.0189	0.1734*
	(0.0608)	(0.0646)	(0.0799)	(0.0699)	(0.0788)	(0.0957)
N	3892	3870	3886	3892	3871	3889
R-Squared	0.0606	0.0312	0.0500	0.0838	0.0807	0.0525
Control Mean	0.0015	-0.0006	-0.0014	-0.0032	-0.0045	-0.0036

Notes: * 0.10 ** 0.05 *** 0.01. Standard errors clustered at mosque-level. Each column is an index created following Anderson (2008) of the respondent's own beliefs (Personal), of what other men in the community think (Norms), and of what Islamic teaching says (Religious), on women's rights (1)-(3) and neighbors' rights (4)-(6).

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Appendix: Table - Heterogeneity on Mosque-Level Dimensions

Table A17: Heterogeneity on Mosque-Level Dimensions

Dependent Variable	Women's Rights		Neighbors' Rights	
	(1)	(2)	(3)	(4)
Women	0.2408***	0.2357	0.1368	-0.1351
	(0.0891)	(0.1527)	(0.1076)	(0.1993)
Neighbors	0.0989	0.2616	0.1094	0.0800
	(0.0939)	(0.1743)	(0.1120)	(0.1999)
Women × Attend. T2	-0.0077		-0.0663	
	(0.1173)		(0.1497)	
Women × Attend. T3	-0.0126		-0.1797	
	(0.1332)		(0.1652)	
Neighbors × Attend. T2	-0.1594		0.0857	
	(0.1358)		(0.1655)	
Neighbors × Attend. T3	-0.0214		0.1538	
	(0.1415)		(0.1582)	
Women × Residential		-0.0026		0.1392
		(0.1099)		(0.1328)
Neighbors × Residential		-0.1646		0.0809
		(0.1168)		(0.1426)
N	3892	3892	3892	3892
R-Squared	0.0319	0.0325	0.0239	0.0193
Control Mean	-0.0003	-0.0003	0.0019	0.0019

Notes: * 0.10 ** 0.05 *** 0.01. Standard errors clustered at mosque-level. Outcomes are the primary behavioral indices, created following Anderson (2008). Columns (1) and (3) interact the treatments with attendance tercile (lowest as base), (2) and (4) with the mosque's residential status.

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Appendix: Table - Beliefs of Imams

Table A18: Beliefs of Imams

Dependent Variable	**Women's Rights Beliefs**	**Neighbors' Rights Beliefs**
	(1)	(2)
Women Secular	0.1404	0.0442
	(0.1083)	(0.1127)
Women Religious	0.2007**	-0.0154
	(0.0967)	(0.1086)
Neighbors Secular	0.2168**	0.0278
	(0.1013)	(0.1021)
Neighbors Religious	0.1993**	0.0041
	(0.0907)	(0.1174)
N	242	242
R-Squared	0.0664	0.0078
Control Mean	-0.0024	0.0038
Ho: Women Rel = Women Sec	0.6186	0.6388
Ho: Neighbors Rel = Neighbors Sec	0.8734	0.8508

Notes: * 0.10 ** 0.05 *** 0.01. Standard errors are robust. Each column is an index of imams' own beliefs created following Anderson (2008).

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Appendix: Table - Domain Expansion and Job Satisfaction of Imams

Table A19: Domain Expansion and Job Satisfaction of Imams

Dependent Variable	Domain Expansion				Job Satisfaction	
	Non-Rel	Rel Politics	Vote Advice	Health Politics	Autonomy	Satisfaction
	(1)	(2)	(3)	(4)	(5)	(6)
Women Secular	0.0632	0.0396	-0.4168	-0.0672	0.6760*	0.4663*
	(0.4316)	(0.4072)	(0.6212)	(0.4030)	(0.3633)	(0.2693)
Women Religious	-0.5943	-0.3306	-0.6422	0.0988	0.2466	0.1078
	(0.4464)	(0.4486)	(0.6183)	(0.3986)	(0.4103)	(0.2957)
Neighbors Secular	0.1497	-0.4576	-1.0383*	-0.4370	0.3464	-0.0421
	(0.4070)	(0.4456)	(0.5380)	(0.4099)	(0.3730)	(0.3252)
Neighbors Religious	-0.0388	-0.0966	-0.3906	-0.5991	-0.0392	-0.1609
	(0.3789)	(0.4134)	(0.5390)	(0.4693)	(0.3430)	(0.2797)
N	241	242	242	242	238	236
R-Squared	0.0169	0.0213	0.0276	0.0167	0.0223	0.0423
Control Mean	7.4865	7.8667	5.5333	7.2533	7.7838	8.4306
Ho: Women Rel = Women Sec	0.2292	0.4850	0.7597	0.7310	0.3366	0.2234
Ho: Neighbors Rel = Neighbors Sec	0.6842	0.4961	0.2809	0.7676	0.3296	0.7208

Notes: * 0.10 ** 0.05 *** 0.01. Respondents are imams. Standard errors are robust. All outcomes are 0-10 ratings: whether imams should have more say in non-religious matters (1), religion a bigger role in politics (2), imams advising congregants on voting (3) and on health (4); satisfaction with job autonomy (5) and with the job overall (6).

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Appendix: Table - Post-Treatment Sermon Content

Table A20: Post-Treatment Sermon Content

Dependent Variable	**Women's Rights**	**Neighbors' Rights**
	(1)	(2)
Women Secular	0.0601	0.1540
	(0.2508)	(0.2358)
Women Religious	0.9400**	0.0949
	(0.3977)	(0.1804)
Neighbors Secular	-0.2834*	1.5363***
	(0.1484)	(0.4584)
Neighbors Religious	-0.0444	0.8891**
	(0.2133)	(0.3589)
N	305	305
R-Squared	0.0866	0.0973
Control Mean	0.4118	0.2157
Ho: Women Rel = Women Sec	0.0407	0.8198
Ho: Neighbors Rel = Neighbors Sec	0.1723	0.2538

Notes: * 0.10 ** 0.05 *** 0.01. Standard errors are robust. Each column is a 0-10 LLM content rating of sermon content 1 week after treatments.

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Appendix: Table - Sermon Content in the Long Run

Table A21: Sermon Content 6-12 Months After the Intervention

Dependent Variable	**Women's Rights**	**Neighbors' Rights**
	(1)	(2)
Women Secular	0.2990	-0.3323***
	(0.3764)	(0.1091)
Women Religious	-0.0315	-0.0512
	(0.3004)	(0.1652)
Neighbors Secular	0.1246	0.1651
	(0.3142)	(0.2354)
Neighbors Religious	0.6972**	0.1585
	(0.3315)	(0.2143)
N	299	299
R-Squared	0.0291	0.0243
Control Mean	0.6176	0.3529
Ho: Women Rel = Women Sec	0.4388	0.0314
Ho: Neighbors Rel = Neighbors Sec	0.1515	0.9810

Notes: * 0.10 ** 0.05 *** 0.01. Standard errors are robust. Each column is a 0–10 LLM content rating of the sermon recorded 6–12 months after the intervention.

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Appendix: Table - Post-Treatment Attendance

Table A22: Post-Treatment Attendance

Dependent Variable	Attendance			
	Log(Attendance)	**Raw**	**Winsor95**	**Winsor90**
	(1)	(2)	(3)	(4)
Women Secular	-0.0435	-36.8757	-3.9741	6.6361
	(0.0828)	(23.5111)	(24.1495)	(22.6651)
Women Religious	-0.2232**	26.7550	-65.1660**	-56.5818**
	(0.1029)	(37.3163)	(28.8019)	(26.2370)
Neighbors Secular	0.0454	33.0315	11.7426	18.7582
	(0.0890)	(30.3928)	(27.5342)	(24.3549)
Neighbors Religious	0.0212	13.5370	10.0373	16.4133
	(0.0818)	(27.5961)	(24.1145)	(22.4438)
N	304	304	304	304
R-Squared	0.6430	0.8594	0.6685	0.6336
Control Mean	5.6059	386.8738	352.8447	331.2039
Ho: Women Rel = Women Sec	0.1093	0.1017	0.0517	0.0345
Ho: Neighbors Rel = Neighbors Sec	0.8031	0.5543	0.9537	0.9317

Notes: * 0.10 ** 0.05 *** 0.01. Standard errors are robust. Outcomes are enumerator-counted attendance 1 week after treatments, in logs and in raw and winsorized counts.

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Appendix: Table - Post-Treatment Attendance (Week 2)

Table A23: Post-Treatment Attendance (Week 2)

Dependent Variable	**Log(Attendance)**	**Raw**	**Winsor99**	**Winsor95**
	(1)	(2)	(3)	(4)
Women Secular	-0.0501	-41.4071**	-39.2622*	-10.0528
	(0.0753)	(20.0444)	(21.0897)	(24.0324)
Women Religious	-0.2181*	14.3745	-47.1840*	-57.0017*
	(0.1144)	(26.4917)	(25.2824)	(30.1634)
Neighbors Secular	0.0381	12.5475	-5.7466	6.9440
	(0.0909)	(30.1507)	(28.3387)	(28.8563)
Neighbors Religious	-0.0798	18.5928	-8.8099	-4.4398
	(0.1152)	(47.3944)	(34.4335)	(29.4922)
N	303	303	303	303
R-Squared	0.5736	0.8271	0.7958	0.6273
Control Mean	5.6059	386.8738	386.8738	352.6117
Ho: Women Rel = Women Sec	0.1532	0.0294	0.7473	0.1482
Ho: Neighbors Rel = Neighbors Sec	0.3522	0.9033	0.9357	0.7433

Notes: * 0.10 ** 0.05 *** 0.01. Standard errors are robust. Outcomes are enumerator-counted attendance on the day of the second treatment sermon (control mosques: two Fridays after the first recording), in logs and in raw and winsorized counts; first-week attendance is a control.

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Appendix: Table - Attendance by Distance to the Nearest Mosque

Table A24: Post-Treatment Attendance by Tercile of Distance to Nearest Mosque

Dependent Variable	**Log(Attendance)**
	(1)
Women Secular	-0.2787**
	(0.1295)
Women Religious	-0.3083
	(0.2069)
Women Secular	0.3248*
× Distance T2	(0.1780)
Women Religious	0.2220
× Distance T2	(0.2412)
Women Secular	0.3179*
× Distance T3	(0.1873)
Women Religious	-0.0706
× Distance T3	(0.2776)
N	293
R-Squared	0.6586
Control Mean	5.5853

Notes: * 0.10 ** 0.05 *** 0.01. Standard errors are robust. Outcome is log attendance 1 week after treatments; women's rights arms interacted with distance-to-nearest-mosque tercile (comparison group: neighbors' rights and control mosques).

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Appendix: Table - Attendance in the Long Run

Table A25: Attendance 6-12 Months After the Intervention

Dependent Variable	Attendance			
	Log(Attendance)	Raw	Winsor99	Winsor95
	(1)	(2)	(3)	(4)
Women Secular	-0.1647	-84.4747**	-77.9997**	-59.9145*
	(0.1040)	(39.4417)	(37.1185)	(33.4537)
Women Religious	-0.2681**	-75.0436	-91.0629**	-78.3110**
	(0.1073)	(47.8537)	(41.3555)	(38.1533)
Neighbors Secular	-0.0099	-13.5755	-16.4623	-8.6367
	(0.0975)	(40.5909)	(38.5924)	(33.9795)
Neighbors Religious	-0.0557	-9.1812	-12.4012	-3.6090
	(0.1019)	(43.4043)	(40.7406)	(37.0018)
N	305	305	305	305
R-Squared	0.4947	0.5116	0.4795	0.4622
Control Mean	5.6644	411.2621	403.7379	382.5728
Ho: Women Rel = Women Sec	0.4106	0.8416	0.7537	0.6410
Ho: Neighbors Rel = Neighbors Sec	0.6875	0.9198	0.9238	0.8964

Notes: * 0.10 ** 0.05 *** 0.01. Standard errors are robust. Outcomes are enumerator-counted attendance at a Friday prayer 6-12 months after the intervention, in logs and in raw and winsorized counts.

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